

JANARDAN SINGH CHAUHAN

Ph.D. · M.Tech. (Foundation Engineering) · M.B.A.

Senior Professor of Civil Engineering & Former Director

Samrat Ashok Technological Institute (SATI), Vidisha, Madhya Pradesh 464 001, India

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PROFILE

Civil engineer and academic leader with four decades across industry, research, teaching, and institutional administration. Senior Professor and former Director (2016–2022) of Samrat Ashok Technological Institute (SATI), Vidisha — one of India's oldest engineering institutions (est. 1960). A pioneer of low-cost, sustainable building materials and cost-effective construction technology in central India: founded and led a HUDCO-approved building technology centre whose methods were nationally recognised and disseminated internationally through UNCHS (UN-Habitat). Supervisor of 13 completed doctoral theses; author of 76+ research papers and 12 books; holder of three granted patents (one UK, two Indian); principal investigator or coordinator of 23+ sponsored projects funded by national agencies and ministries.

RESEARCH INTERESTS

Foundation engineering — soil-structure interaction and foundations under vibratory loading

Alternative building materials — low-cost, waste-derived and energy-efficient materials; cost-effective construction techniques

Sustainable & green buildings — thermal performance, energy-efficient design, green construction concepts

Construction management — project economics, quality assurance, TQM in construction and education

Trenchless technology — subsurface infrastructure development and no-dig methods

Green hydrogen — emerging interest — hydrogen in the built environment and infrastructure

EDUCATION

Ph.D., Civil Engineering · Barkatullah University, Bhopal 1999

Thesis: "Comparative Study on Techno-Economic Aspects & Thermal Performance of Experimental and Conventional Housing Construction Techniques"

M.B.A. (Consortium Programme) · Devi Ahilya Vishwavidyalaya, Indore 1997

First Division with Honours — 90.0%

M.Tech., Civil Engineering — Foundation Engineering (Applied to Vibratory Systems) · Barkatullah University, Bhopal 1996

First Division — 72.6%

B.E., Civil Engineering · Barkatullah University, Bhopal 1991

First Division with Honours — 81.2%

Diploma in Civil Engineering · M.P. Board of Technical Education, Bhopal
First Division with Honours — 84.4%

1983

ACADEMIC & PROFESSIONAL APPOINTMENTS

Senior Professor, Department of Civil Engineering, SATI, Vidisha Feb 2022 – present
Senior-most professorial grade; research, doctoral supervision and postgraduate teaching in civil engineering.

Director (Head of Institution), SATI, Vidisha Aug 2016 – Feb 2022
Chief executive of an autonomous, NAAC-accredited engineering institution (est. 1960; foundation stone laid by Prime Minister Jawaharlal Nehru; alumni include Nobel Peace Laureate Kailash Satyarthi). Institute was selected under the World Bank-assisted TEQIP-III programme (₹15 crore) during this tenure.

Vice-Principal & Officiating Director, SATI, Vidisha Aug 2015 – Aug 2016
Institution-level administration; period of the institute's NAAC accreditation (March 2016).

Head, Department of Civil Engineering, SATI, Vidisha Sep 2002 – Aug 2016
Led the department's teaching, laboratories, sponsored research and industry consultancy for fourteen years.

Head, Department of Management, SATI, Vidisha Jul 2001 – Oct 2006
Concurrent charge; built the institute's management programme.

Reader, Department of Civil Engineering, SATI, Vidisha 1995 – 2000

Project Engineer & Lecturer, SATI, Vidisha 1992 – 1995
Anchored the HUDCO-approved Civil Engineering Technology Development Centre (CETDEC) from its early years.

Assistant Engineer, M/s Ankur Construction Company, Vidisha 1985 – 1992
Seven years of field construction practice prior to academia.

CENTRE LEADERSHIP — CETDEC

Founded in 1991 on the SATI campus as a HUDCO-approved institutional building centre, the Civil Engineering Technology Development Centre (CETDEC) demonstrates and propagates experimental housing using low-cost, innovative construction techniques and alternative building materials. Its work attracted a KfW (German) grant of ₹12.60 lakh (1995) and a Mobile Building Centre grant of ₹13.13 lakh (1999) through HUDCO, and it has executed construction worth ₹1.5 crore for the district administration. Dr. Chauhan has anchored the Centre since 1992 and serves as its Secretary.

RESEARCH SUPERVISION (AS OF 2023)

Ph.D. theses — 13 awarded · 3 in progress

M.Tech. theses — 132 guided

M.B.A. dissertations — 52 guided

PATENTS

United Kingdom patent — 1 granted

Indian patents — 2 granted

Patent applications — 2 filed (India)

Patent numbers, titles and grant dates available on request.

SPONSORED RESEARCH, DEVELOPMENT & CONSULTANCY PROJECTS

1. **Establishment of the Building Centre (CETDEC).** HUDCO, New Delhi; Project Engineer. 1992-93 · ₹5.00 lakh.
2. **KfW grant for experimental low-cost housing technology.** HUDCO — KfW (German) Grant; Coordinator. 1995-96 · ₹12.60 lakh.
3. **Experimental construction — F-type quarters (planning, supervision & construction).** National Buildings Organisation, New Delhi; Lead. 1995-96 · ₹16.00 lakh.
4. **Major Research Project — Comparative study of thermal performance of experimental & conventional housing techniques.** University Grants Commission, New Delhi; Coordinator. 1995-96 · ₹3.70 lakh.
5. **Development of low-cost construction building materials & technologies.** AICTE, New Delhi (TAPTEC); Coordinator. 1996-97 · ₹7.00 lakh.
6. **Waste management, effluent treatment and recycling.** AICTE, New Delhi (TAPTEC); Coordinator. 1996-97 · ₹4.00 lakh.
7. **Experimental construction — “Kabir Kutir” demonstration housing.** HUDCO, New Delhi; Lead. 1997-98 · ₹22.00 lakh.
8. **Sericulture sheds — experimental construction.** District Collector, Vidisha; Lead. 1997-98 · ₹15.00 lakh.
9. **Primary school buildings — experimental construction.** Rajiv Gandhi Shiksha Mission; Lead. 1997-98 · ₹15.00 lakh.
10. **Computer-aided design in civil engineering.** AICTE, New Delhi (TAPTEC); Principal Investigator. 1999-2000 · ₹8.00 lakh.
11. **Mobile Nirmiti Kendra (mobile building centre).** Ministry of Urban Affairs, GoI; Coordinator. 1999-2000 · ₹13.13 lakh.
12. **Rural Building Centre.** Ministry of Rural Development, GoI; Coordinator. 2001-02 · ₹12.00 lakh.
13. **Technology Park.** HUDCO, New Delhi; Coordinator. 2001-02 · ₹7.50 lakh.
14. **Rural housing & habitat development.** Ministry of Rural Development, GoI (Innovative Stream); Coordinator. 2001-02 · ₹10.50 lakh.
15. **Energy Park.** Ministry of Non-Conventional Energy Sources, GoI; Coordinator. 2002-03 · ₹5.00 lakh.
16. **Major Research Project — Energy-efficient building.** University Grants Commission, New Delhi; Investigator. 2002-03 · ₹5.08 lakh.
17. **Model Basti (model settlement).** Ministry of Rural Development, GoI; Coordinator. 2002-03 · ₹8.00 lakh.
18. **Modernisation of the Environmental Engineering Laboratory.** AICTE, New Delhi (MODROB); Coordinator. 2003-04 · ₹8.00 lakh.
19. **National Habitat Centre on Subsurface & Trenchless Technology.** AICTE, New Delhi (NAFETIC); Coordinator. 2005-06 · ₹15.00 lakh (total project cost ₹1.08 crore).
20. **International conference support & development of the CETDEC laboratory.** AICTE, New Delhi; Coordinator. 2007-08 · ₹3.00 lakh.

21. **Modernisation of the Civil Engineering Laboratory.** AICTE, New Delhi (MODROB); Coordinator. 2007-08 · ₹8.50 lakh.
22. **National Retrofitting Clinic.** AICTE, New Delhi (NCP); Coordinator. 2008-09 · ₹30.00 lakh.
23. **Third-Party Quality Assurance (TPQA) consultancy for public works.** Central Public Works Department, Bhopal; Lead Consultant. 2015 · ₹1.50 crore.

Representative list; additional smaller projects omitted. Aggregate sponsored funding above ₹3.8 crore across 23+ projects from 12+ national agencies (HUDCO, AICTE, UGC, NBO, CPWD, and Ministries of Rural Development, Urban Affairs and Non-Conventional Energy).

PUBLICATIONS

Journal Articles

1. Zaidi, S. A. M., Chauhan, J. S., Sharma, P., & Amritphale, S. S. "Preparation of geopolymer cement with dry activators by one-part system." *Journal of Emerging Technologies and Innovative Research*, 8(8), ISSN 2349-5162, 2021.
2. Patel, Y. S., Chaturvedy, G. K., Kumar, A., & Chauhan, J. S. "Solid waste management through different methods." *International Journal for Scientific Research & Development*, 4(3), 1430-1433, 2016.
3. Patel, Y. S., Chaturvedy, G. K., Kumar, A., & Chauhan, J. S. "A review paper on study of solid waste management through different methods." *International Journal for Scientific Research & Development*, 4(3), 1434-1436, 2016.
4. Saxena, S., Ramakrishnan, N., & Chauhan, J. S. "Assessing fracture behaviour of circumferentially through-wall-cracked elbows subjected to closing and opening moments." *Nuclear Engineering and Design (Elsevier)*, 239(1), 9-15, 2009.
5. Saxena, S., Ramakrishnan, N., & Chauhan, J. S. "Fatigue life prediction analysis of surface-cracked straight pipes." *Transactions of the Indian Institute of Metals*, 2009.
6. Panthi, S. K., Ramakrishnan, N., Das Gupta, R., & Chauhan, J. S. "Study of effect of load on springback in sheet metal bending." *Transactions of the Indian Institute of Metals*, 61(1), 1-5, 2008.
7. Shrivastava, N. K., Chauhan, J. S., & Gumastha, L. K. "Cost-effective infrastructure development through trenchless technology." *International Journal of Trenchless Technology (CIDC)*, 2008.
8. Panthi, S. K., Ramakrishnan, N., Pathak, K. K., & Chauhan, J. S. "An analysis of springback in sheet metal bending using finite element method (FEM)." *Journal of Materials Processing Technology (Elsevier)*, 186, 120-124, 2007.
9. Panthi, S. K., Ramakrishnan, N., Pathak, K. K., & Chauhan, J. S. "Prediction of springback in straight flanging using finite element method." *CMC: Computers, Materials & Continua (Tech Science Press)*, 6(1), 13-19, 2007.
10. Saxena, S., Ramakrishnan, N., & Chauhan, J. S. "Prediction of smooth elbow behaviour using finite element method." *Transactions of the Indian Institute of Metals*, 60(6), 553-558, 2007.
11. Panthi, S. K., Pathak, K. K., Chauhan, J. S., & Ramakrishnan, N. "Application of neural network in straight flanging process." *Metals Materials and Processes (Meshap Science Publishers)*, 18(2), 191-200, 2005.

12. Shrivastava, N. K., Chauhan, J. S., & Gumastha, L. K. "Recent trends in computer-aided cost optimisation in rural housing." *Journal of the Institution of Engineers (India); proceedings pub. Nov 2003, p. 226, 2004.*
13. Shrivastava, N. K., Chauhan, J. S., & Gumastha, L. K. "A review of computer-aided design of rural roads as per PMGSY." *Journal of Urban Street for Transportation, New Delhi; proceedings p. 267, 2004.*
14. Chauhan, J. S., Shrivastava, N. K., & Gumastha, L. K. "Computer-aided design of cost-effective earthquake-resistant rural housing." *Journal of the Indian Buildings Congress, New Delhi; proceedings p. 57, 2004.*
15. Shrivastava, N. K., Chauhan, J. S., & Gumastha, L. K. "Computer-aided analysis of cost-effective construction techniques in rural roads." *Journal of the Indian Roads Congress, New Delhi (accepted), 2004.*

International Conference Papers

1. Saxena, S., Ramakrishnan, N., & Chauhan, J. S. "EPFM-based predictions of important experimental results in circumferentially through-wall-cracked elbows." *5th Int. Conf. on Creep, Fatigue and Creep-Fatigue Interaction, IGCAR, Kalpakkam, India, 2008.*
2. Chauhan, J. S. "Prospects of e-governance / Principles of total quality management (two papers)." *6th Global Conference on Flexible Systems Management (GLOGIFT-06), Asian Institute of Technology, Bangkok, Thailand, 2006.*
3. Chauhan, J. S., & Katiyar, S. K. "GPS technology applications in the construction industry." *World Conf. on IT Solutions for the Design & Management of Infrastructure Construction Projects, India Habitat Centre, New Delhi, 2006.*
4. Chauhan, J. S., & Malviya, R. K. "Energy-efficient and green buildings technology — an overview." *Int. Seminar on Retrofitting & Buildings to Withstand Earthquake and Green Construction Concept, M.P. Academy of Administration, Bhopal, 2006.*
5. Chauhan, J. S. "Cost-effective, environment-friendly housing through planning & design." *Int. Seminar on Retrofitting & Buildings to Withstand Earthquake and Green Construction Concept, Bhopal, 2006.*
6. Chauhan, J. S. "Trenchless technology — a viable solution for Indian conditions." *Int. Conf. on New Materials, Devices and NO-DIG Technologies in Water Supply & Sewage Systems, Kielce University of Technology, Poland, 2005.*
7. Chauhan, J. S. "Alternate energy-efficient building materials for low-cost housing." *Int. Conf. on Intelligent, Green & Energy-Efficient Building, Beijing, China, 2005.*
8. Panthi, S. K., Pathak, K. K., Chauhan, J. S., & Ramakrishnan, N. "An analysis of springback in sheet metal bending using FEM." *Int. Conf. on Computational & Experimental Engineering and Sciences (ICCES-05), IIT Madras, Chennai, 2005.*
9. Chauhan, J. S. "Geoinformatics techniques in power development and management (with two further papers: TQM; emerging trends in civil engineering construction)." *Int. Conf. on Management of IT & Infrastructure for Flexibility & Competitiveness, RGPV, Bhopal, 2005.*
10. Chauhan, J. S. "Quality management in distance education." *ICDE International Conference on Open Learning & Distance Education, IGNOU, New Delhi, 2005.*
11. Chauhan, J. S. "Subsurface infrastructure development & management — a case study of India." *22nd International NO-DIG Conference & Exhibition, Hamburg, Germany, 2004.*
12. Chauhan, J. S. "Total quality management in HRM." *4th Global Conference on Flexible Systems Management, Mussoorie, India, 2004.*

13. Chauhan, J. S. "Benchmarking in TQM for excellence." *Int. Conf. on Construction Opportunities & Strategies for Action — Asia-Pacific, Middle East & African Regions, CIDC, New Delhi, 2004.*
14. Chauhan, J. S. "Technology, environment & habitat development." *International Seminar on Habitat Development, Colombo, Sri Lanka, 2004.*
15. Chauhan, J. S. "Wind engineering & science." *International Workshop on Wind Engineering & Science (WES-04), IIT Roorkee, at India International Centre, New Delhi, 2004.*
16. Chauhan, J. S., & Shrivastava, N. K. "Computer-aided analysis of cost-effective construction techniques in rural housing." *XXXI IAHS World Congress on Housing, Concordia University, Montreal, Canada, 2003.*

National Conferences, Seminars & Workshops

1. Chauhan, J. S. "Building an ecologically sustainable society." *Indian Social Science Congress, 2013.*
2. Chauhan, J. S. "Scientific solutions for waste management." *National Seminar on Environment & Development, EPCO, Bhopal, 2006.*
3. Chauhan, J. S. "Construction guarantee fund." *Seminar organised by CIDC, supported by National Science Foundation (USA), BSI (UK) & Singapore International Arbitration Centre, New Delhi, 2005.*
4. Chauhan, J. S. "Role of building centres in HRD; Cost reduction in buildings; Role of GPS in infrastructure development (three papers)." *8th Convention of Construction — Seminar on Infrastructure Development in M.P., CIDC, New Delhi, 2005.*
5. Chauhan, J. S. "Energy efficiency in buildings." *Conf. on Emerging Opportunities for Consulting Service Providers in the Context of Globalisation, International Council of Consultants, New Delhi, 2005.*
6. Chauhan, J. S. "Sustainable development in the housing sector." *National Workshop on Strategy for New State Housing Policy, Institute of Town Planners India, Bhopal, 2004.*
7. Chauhan, J. S., Murthy, A. G. K., & Punjabi, K. K. "Waste utilisation and management." *National Workshop on Scientific Solutions for Waste Management & Pollution Control, SATI, Vidisha, 2004.*
8. Chauhan, J. S., & Joshi, Y. P. "Benefits of steel-fibre-reinforced concrete for various constructions." *Seminar on Recent Trends in Building Materials, MRSI & Regional Research Laboratory (CSIR), Bhopal, 219-225, 2004.*
9. Chauhan, J. S., Shrivastava, N. K., & Gumastha, L. K. "Alternative building materials and components for rural housing." *Seminar on Recent Trends in Building Materials, MRSI & RRL (CSIR), Bhopal, 316-321, 2004.*
10. Chauhan, J. S. "Tube and dug-well recharging as an effective strategy for water management in water-scarce regions — a case study of Vidisha district." *National Seminar on Ground Water Conservation & Recharging in M.P., PHED M.P. & UNICEF, Bhopal, 2003.*
11. Chauhan, J. S. "Strengthening of building centres." *National Induction Training Programme on New Building Materials & Cost-Effective Construction Techniques, SATI, Vidisha, 2001.*
12. Chauhan, J. S. "Lignin recovery from liquid waste of bagasse alkali pulping; Cost-effective building materials from industrial wastes; Waste management — an overview (three papers)." *National Seminar on Waste Management, SATI, Vidisha, 1999.*

13. Chauhan, J. S. "Strengthening of building centres." *Two-day Workshop for Development of Building Centres, HUDCO, Mumbai*, 1999.
14. Chauhan, J. S. "Cost-effective materials from industrial wastes." *Conf. on Industrial Wastes in Development of Building Materials, Regional Research Laboratory (CSIR), Bhopal*, 1998.
15. Chauhan, J. S. "Use of ferrocement techniques." *Training Programme for Polytechnic Teachers, SATI, Vidisha*, 1996.
16. Chauhan, J. S. "Low-cost construction." *Training Programme of Project Managers, HUDCO, New Delhi*, 1996.
17. Chauhan, J. S. "Innovative low-cost building materials from industrial wastes; Project cost control by inventory management techniques (two papers)." *National Seminar on Role of Engineering Institutions, Construction Industry & Building Research Institutes in Low-Cost Construction, MITS, Gwalior*, 1996.
18. Chauhan, J. S. "Education and economy." *Seminar on Quality Assurance in Technical Education — Challenges of the New Economic Policy, SATI, Vidisha*, 1995.
19. Chauhan, J. S. "Activities of the Civil Engineering Technology Development Centre, Vidisha." *Seminar on the Role of Building Centres in Low-Cost Housing, HUDCO & M.P. Avas Vikas Sansthan, Bhopal*, 1995.
20. Chauhan, J. S. "Alternate building materials & technologies." *Winter School on Alternative Building Materials & Cost-Effective Technologies, SATI, Vidisha*, 1994.
21. Chauhan, J. S. "Stabilised mud blocks with fibres; Cost comparison — alternate and conventional housing materials (two papers)." *Seminar on Alternate Housing Materials, Regional Research Laboratory (CSIR), Bhopal*, 1993.
22. Chauhan, J. S. "Need for technological developments in the rural sector." *Workshop on Rural Technology, Vigyan Sabha, Bilaspur*, 1992.
23. Chauhan, J. S. "Economical sanitation in rural areas; Application of polymers & plastics in buildings; Cost reduction in rural housing (three papers)." *Seminar-cum-Integrated Training Programme on Low-Cost Building Materials & Housing, SATI Vidisha & CBRI Roorkee*, 1991.

Professional & Trade Publications

1. Chauhan, J. S. "Space planning for thermal comfort house." *Nirman Sarika (special issue), BMTPC, New Delhi*, 4(3), 2015.
2. Chauhan, J. S. "Planning for thermal comfort building." *NCHF Bulletin, Vol. XXVIII, National Cooperative Housing Federation, New Delhi*, 2015.
3. Chauhan, J. S. "Energy-efficient building for urban areas." *NCHF Bulletin, Vol. XXVI*, 2013.
4. Chauhan, J. S. "Planning & design of cost-effective & environment-friendly houses in urban areas." *Nirman Sarika (special issue), BMTPC, New Delhi*, 2013.
5. Chauhan, J. S. "Design parameters for energy-efficient buildings — consulting ahead." *Journal of the Consultancy Development Centre, Vol. 4*, 2010.
6. Chauhan, J. S. "Quality, durability, performance & service life of building materials." *Nirman Sarika, BMTPC World Habitat Day special issue*, 2009.
7. Shrivastava, N. K., & Chauhan, J. S. "Computer-aided design of cost-effective earthquake-resistant group housing in semi-urban areas." *Nirman Udyog (CIDC)*, 1(2), 4-9, 2008.

8. Malviya, R. K., & Chauhan, J. S. "Energy efficiency in buildings — effect of design parameters." *Nirman Udyog (CIDC)*, 2008.
9. Bhangale, P. P., & Chauhan, J. S. "Characterisation of ponded fly ash use in concrete." *Nirman Udyog (CIDC)*, 2008.
10. Chauhan, J. S., Sharma, D. K., & Singh, M. "Prospects of e-governance." *Nirman Udyog (CIDC)*, 2008.
11. Chauhan, J. S. "A unique feature of SATI." *International Journal of the SATI International Alumni Association*, 2004.
12. Chauhan, J. S. "Kam laagat mein bhavan nirman — sapna nahin, ek sachai (Hindi)." *Udyamita Samachar Patra, Bhopal*, 1997.

Books

12 books published; 2 in progress (as of 2023).

Author/subject-matter editor of a 12-volume technology-dissemination pocket-book series published by HUDCO, New Delhi, for grassroots construction workers — graphical guides to twelve cost-effective technologies (filler slabs; brick panels & joists; ferrocement roofing channels & doors; rat-trap bond masonry; rammed-earth walls; arches & vaults; arch foundations; micro-concrete roofing tiles; water-storage tanks; cement-concrete/stone blocks). Planned for worldwide circulation through UNCHS (UN-Habitat).

HONOURS & AWARDS

National Recognition & Silver Plaque, HUDCO Silver Jubilee	1995
<i>for outstanding work in low-cost housing by the Building Centre led by Dr. Chauhan; presented by the Hon'ble Union Minister for Urban Affairs & Employment, New Delhi</i>	
Certificate of Excellence & Gold Medal, Friendship Forum of India, New Delhi	2008
<i>for outstanding achievements in low-cost civil engineering and services promoting India-international cooperation</i>	
Fellow, Temasek Foundation-NUS Programme for Leadership in University Management	2015
<i>National University of Singapore</i>	
Building Centre Achievement Award, Shelter Group, Calcutta	1994
<i>for the achievements of the CETDEC building centre</i>	
Annual Award for Extraordinary Performance, Director, SATI	1992-2004
<i>conferred every year from 1992 to 2004</i>	
Second-Best Exhibition Stall, CSIR National Workshop on Rural Technologies, Bhopal	1996
<i>exhibition inaugurated by the Hon'ble Chief Minister of Madhya Pradesh</i>	

CONFERENCES & TRAINING PROGRAMMES ORGANISED (SELECTED)

International Conference on Management of IT & Infrastructure Development for Flexibility & Competitiveness — <i>RGPV, Bhopal</i>	2005
International Seminar on Retrofitting & Buildings to Withstand Earthquake and Green Construction Concept — <i>M.P. Academy of Administration, Bhopal (with IndSTT & EPCO)</i>	2006

National Seminar on Waste Management — <i>SATI, Vidisha</i>	1999
National Seminar on Energy-Efficient Buildings — <i>Dept. of Civil Engineering, SATI</i>	2004
National Seminar on Awareness of Sub-surface & Trenchless Technology — <i>CETDEC, SATI</i>	2004
National Workshop on Scientific Solutions for Waste Management & Pollution Control — <i>SATI, Vidisha</i>	2004
National Seminar on Infrastructure Development in Madhya Pradesh & Chhattisgarh — <i>RGPV, Bhopal</i>	2005
National Conference on Sustainable & Durable Options for Pavements — Quality Control for Road Construction — <i>CETDEC, at Hotel Lake View Ashok, Bhopal</i>	2010
National-level symposium “Anveshan-11” — <i>Dept. of Civil Engineering, SATI</i>	2011
Workshop on Water Conservation & Rehabilitation of Rivers in India (with Magsaysay laureate Dr. Rajendra Singh, the “Waterman of India”) — <i>SATI, Vidisha</i>	2016
Staff development programmes for engineering teachers — new building materials (2001, 2004), intelligent & green building materials (2009), cost-effective construction (2010), economy in sustainable construction (2013) — <i>SATI, Vidisha</i>	2001-13
Two-tier re-skilling programmes for masons, artisans, engineers & building-centre project managers on alternative construction technologies; annual self-employment training for rural & urban youth — <i>CETDEC, SATI</i>	1996-2010
Three-day training programme on remote sensing, GIS & surveying; seminar with Central Water Commission — <i>SATI, Vidisha</i>	2016

INTERNATIONAL ENGAGEMENTS

Imperial College London — collaboration discussions with civil engineering faculty	2004
Glasgow Caledonian University, UK — meetings on a job-oriented training programme for civil engineering students	2004
WRc (Water Research Centre), Swindon, UK — collaborative programme on sewerage & water supply	2004
National University of Singapore — Temasek Foundation-NUS leadership fellowship	2015
Kielce University of Technology, Poland — invited participation, international trenchless-technology conferences	2005, 2006
Professional engagements across 13 countries: Germany, United Kingdom, France, Switzerland, Italy, Sweden, Finland, Poland, China, Singapore, Thailand, Sri Lanka, and India.	

PROFESSIONAL MEMBERSHIPS

- Life Member, Indian Society for Technical Education (ISTE), New Delhi — 1993
- Life Member, The Institution of Engineers (India) — 2002
- Life Member, International Society for Rock Mechanics — 2002
- Life Member, Indian Roads Congress — 2004
- Life Member, Indian Society for Trenchless Technology (IndSTT) — 2004
- Life Member, Indian Society for Wind Engineering — 1998
- Life Member, Institute of Urban Transport (India) — 2001
- Life Member, Indian Science Congress Association, Kolkata — 2003
- Life Member, Association for Environmental Management — 1996

- Life Member, International Council of Consultants (ICC) — 2004
- Life Member, Council for Application of Actuarial Science in Risk Management (CARM) — 2004
- Life Member, All India Housing Development Association, New Delhi — 2004
- Life Member, Global Institute of Flexible Systems Management (GIFT), New Delhi — 2004
- Life Member, Consultancy Development Centre (promoted by DSIR, Ministry of Science & Technology) — 2006
- Life Member, Society for Management & Development Studies, Bhopal
- Life Member, SATI International Alumni Association — 2004

ADVISORY & COMMUNITY SERVICE (SELECTED)

- Secretary, Civil Engineering Technology Development Centre (CETDEC), Vidisha
 - Member, State-Level Monitoring Committee for Building Centres, Government of Madhya Pradesh
 - Technical Advisor, Srijan Building Centre, Central Jail, Bhopal (construction-skills rehabilitation programme)
 - Technical Advisor, Rajiv Gandhi Shiksha Mission & Rajiv Gandhi Watershed Mission, District Vidisha
 - Member, Advisory Board, Entrepreneurship Development Cell, SATI, Vidisha
 - District Coordinator, Madhya Pradesh Vigyan Sabha, Vidisha (1996-97)
 - Life Member, Indian Red Cross Society (1993)
 - Member, Ramkrishna Memorial Education Trust, Vidisha; Patron, Vidisha Pratibha Vikas Sansthan
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References and supporting documentation available on request.